Finalized Project Proposal

Hospital Readmission Risk Predictor: A Phased MLOps Build

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Roles This Week: Team Lead, Documentation Lead, and Communication Lead

Team Name: Epsilon

GitHub Repository:

https://github.com/AngieLie/AngelineSetiawan-hospital-readmission-risk-predictor

Background & Question

Research Question: Can a patient's risk of being readmitted to the hospital within 30 days of discharge be predicted from clinical encounter data, and can that prediction be delivered through a live, self-updating machine learning system rather than a one-off model?

This question matters because hospitals in the United States are financially penalized under the Centers for Medicare & Medicaid Services' (CMS) Hospital Readmissions Reduction Program (HRRP) when their 30-day readmission rates exceed expected benchmarks. The program was designed to tie Medicare payment to the quality of care patients receive after discharge, and it currently affects the majority of eligible U.S. hospitals in some form, with roughly one in fourteen hospitals losing at least one percent of Medicare reimbursement and payment reductions capped at three percent (Centers for Medicare & Medicaid Services, n.d.). A reliable, deployable risk model gives hospitals a way to flag high-risk patients before discharge so that care teams can intervene early — for example, through follow-up calls, care coordination, or medication review — rather than reacting after a readmission has already occurred.

Most classroom-style readmission models stop at reporting an accuracy score. This project instead treats the prediction as one component of a working system: one that logs experiments, promotes the best-performing model, serves predictions through a live API, and, in a stretch phase, retrains and redeploys itself automatically. That operational layer is the real-world niche this project addresses — proving that a model works is different from proving that it can be run responsibly, and repeatedly, in production. The primary stakeholders and intended audience are hospital administrators and care management or utilization-review teams who are accountable for readmission rates and would act on the model's risk scores; a secondary audience is health-system data and ML engineering teams interested in the underlying MLOps pattern itself.

Thirty-day readmission prediction on this population has been studied extensively. The dataset used here was originally assembled and analyzed by Strack et al. (2014), who used a comparable extract of roughly 70,000 diabetic inpatient encounters to study the relationship between HbA1c testing and early readmission, and it has since become a standard benchmark for readmission modeling with logistic regression, random forests, and gradient boosting. This project builds on that prior work rather than trying to out-perform it on accuracy alone; the point of differentiation is the surrounding system — experiment tracking, a model registry, a serving layer, and conditional automated retraining — which most prior academic treatments of this dataset do not include.

Hypothesis: Patients with more complex or intensive care histories — longer hospital stays, more prior inpatient or emergency visits, and a higher number of medications — are more likely to be readmitted within 30 days, because these factors reflect greater underlying disease severity and more complicated post-discharge care needs.

Prediction: A supervised classification model trained on this dataset will identify patients at elevated 30-day readmission risk with meaningfully better performance than a naive baseline (e.g., always predicting the majority class), and utilization-related features such as number of prior inpatient visits, time in hospital, and number of medications will rank among the most important predictors.

Data & Methods

The primary dataset is the Diabetes 130-US Hospitals for Years 1999–2008 dataset, hosted by the UCI Machine Learning Repository (Clore et al., 2014). It covers ten years of clinical care at 130 U.S. hospitals and integrated delivery networks, with approximately 101,766 inpatient diabetic encounters and more than 50 fields per encounter, including demographics, admission and discharge codes, diagnosis codes, medication counts, lab and procedure counts, and prior-year utilization counts. It is licensed CC BY 4.0, requires no login or credentialing, and downloads directly as a CSV, or through the ucimlrepo Python package. The dataset is appropriate for this project because it is large enough to support both a supervised classifier and two required unsupervised analyses, and because it includes a genuine 30-day readmission outcome label together with the demographic, diagnostic, medication, and prior-utilization fields needed to test the stated hypothesis. It will be obtained through a direct CSV download from the UCI repository, so no API key, web scraping, or database access is required.

Preprocessing completed or planned so far includes collapsing the dataset's native three-class outcome (readmitted in under 30 days, readmitted in over 30 days, or not readmitted) into a binary target, standardizing the dataset's missing-value placeholders (recorded as "?" in several fields) into proper null values, and taking an initial inventory of the high-cardinality ICD-9 diagnosis fields to plan a grouping strategy before encoding. For another analyst to reproduce this work, the repository will include the raw CSV alongside a documented, version-controlled preprocessing script rather than an ad hoc notebook, so that the same cleaned table can be regenerated from the original download at any point in the pipeline.

Dataset Currency and Limitations — Justification

Before committing to this dataset, alternatives were considered, including more recently collected clinical data. In practice, no publicly available, modern equivalent exists that matches this dataset's combination of accessibility, sample size, and label quality. Contemporary hospital EHR extracts with real readmission outcomes are protected health information under HIPAA and are not released publicly. The most comparable modern research-grade alternative, the MIMIC-IV critical care database, requires a completed human-subjects research training course, a signed data use agreement, and PhysioNet credentialing before access is granted — a process that is not achievable within this course's timeframe. Synthetic modern datasets (e.g., Synthea-generated records) avoid privacy restrictions but do not carry real-world readmission labels, which would undermine the project's predictive validity. Given these constraints, the Diabetes 130-US Hospitals dataset remains the most defensible choice available: it is genuinely public, requires no gatekeeping, and — unlike a synthetic alternative — reflects real clinical outcomes.

That choice comes with acknowledged limitations that this proposal does not want to understate. The encounters were collected between 1999 and 2008, which predates the HRRP itself (established by the Affordable Care Act and first enforced in fiscal year 2013), predates widespread electronic health record adoption under the Meaningful Use program, and reflects an era of diagnosis coding (ICD-9) and diabetes management practice that has since evolved. Any relationships the model learns describe historical patterns in this population and should be read as a proof of concept for the modeling and MLOps pipeline rather than as a claim about current hospital performance; a model intended for real deployment would need retraining on contemporary, hospital-specific data before use. The dataset is also structurally limited in a way that directly affects validation design: it does not provide an exact admission or discharge calendar date, and it does not expose which of the 130 hospitals, or which state or region, a given encounter came from. Only an anonymized encounter_id, a repeatable patient_nbr, and coded categorical fields (such as admission source and payer code) are available. That constraint shapes the time- and geography-aware validation strategy described in the Analysis Plan below, rather than being treated as a detail to work around silently.

The size of this dataset is a further limitation that is stated directly here rather than left implicit. At roughly 101,766 total encounters, drawn from a single condition population (diabetic inpatients) across 130 hospitals, this dataset is small relative to modern, nationally representative alternatives. For comparison, the AHRQ Healthcare Cost and Utilization Project's Nationwide Readmissions Database (NRD) contains on the order of 16 to 18 million discharges per year across all conditions and a broader hospital sample (Agency for Healthcare Research and Quality, 2024). After collapsing the outcome into a binary 30-day readmission label, the positive class shrinks further, to roughly 11 percent of encounters, or on the order of 11,000 records, and the time-aware, patient-grouped split described in the Analysis Plan reduces the usable training partition further still. This project does not present its data as a large-scale or nationally representative sample, and its results should be read accordingly: as a proof of concept for the modeling and MLOps pipeline built on a real but modest dataset, not as a claim generalizable to the broader U.S. inpatient population. Two design choices follow directly from this acknowledgment. First, model selection favors algorithms that perform reasonably at this scale

— logistic regression, random forest, and gradient-boosted trees — rather than data-hungry deep learning approaches that this dataset is too small to support responsibly. Second, the high-cardinality diagnosis fields will be grouped into a small number of clinically meaningful categories before encoding specifically to avoid overfitting a categorical feature space that would otherwise be large relative to the number of available rows. Access to a modern, larger database such as the NRD requires a paid data use agreement and is not obtainable within this course's scope; that gatekeeping, alongside the licensing and credentialing barriers described above, is part of why the smaller, freely available dataset was retained despite this tradeoff, with the tradeoff itself treated as a limitation to design around rather than a weakness to hide.

Analysis Plan

Additional cleaning and preprocessing

- Group the high-cardinality ICD-9 diagnosis codes (diag_1, diag_2, diag_3) into a smaller number of clinically meaningful categories before encoding, rather than one-hot encoding hundreds of raw codes.
- Encode remaining categorical fields (admission type, discharge disposition, admission source, age bracket) and address class imbalance in the binary target, likely through class weighting first, with resampling (e.g., SMOTE) as a comparison rather than a default.
- Explicitly drop or mask any field that would only be known after discharge, to avoid leaking post-outcome information into the predictors.

Exploratory analysis

- Univariate and bivariate summaries of utilization and demographic fields against the binary readmission target.
- K-means or hierarchical clustering as the first unsupervised model, to group patients into risk profiles based on utilization and demographic features.
- PCA (or t-SNE) as the second unsupervised model, to reduce the 50+ features to a handful of components for visualization and, potentially, as engineered inputs to the supervised model.

Modeling approach

- Baseline: logistic regression, chosen as the simplest reasonable and most interpretable classifier.
- Comparison model 1: random forest, to capture nonlinear interactions among utilization and demographic features.

- Comparison model 2: XGBoost, as the primary candidate for the model ultimately promoted to "Production" in the model registry.

Accounting for time structure (no leakage)

- Because the public release of this dataset does not include an exact calendar date, encounter_id will be used as an ordinal proxy for chronological order (encounters were extracted from the source system in approximately the order they occurred). Splitting will follow a time-aware design: the earliest ~75-80% of encounters by this ordering will form the training set, and the most recent ~20-25% will be held out as a test set — never a random shuffled split, which would let information from later encounters leak into training.
- Because patient_nbr repeats across a patient's multiple encounters, splits will also be grouped at the patient level (e.g., GroupKFold-style logic layered on top of the time-based cut) so that no single patient's records appear in both the training and test partitions.
- Lagging is preserved rather than invented: fields such as number_outpatient, number_emergency, and number_inpatient already describe utilization in the year preceding the index encounter, so they are used as-is as backward-looking features rather than being extended with information that would only be available afterward.
- An engineered "time elapsed since prior encounter" feature will be built strictly from a patient's own prior encounter history (using encounter_order as the time proxy), never from encounters that occur after the one being predicted.
- The same logic is applied to geography: because the public data does not identify which of the 130 hospitals or which state a given encounter belongs to, a true hold-one-hospital-out or hold-one-region-out split is not achievable with this dataset. This is documented as a known limitation rather than approximated with an unsupported proxy; if a location-adjacent coded field (e.g., payer_code) is explored, it will be treated only as a coarse, low-confidence signal and not used to infer or leak hospital identity across the split.

Evaluation and validation

- Because 30-day readmissions are a minority outcome, evaluation will lean on AUC-ROC and recall/precision for the positive (readmitted) class rather than raw accuracy, which can look strong while missing most true readmissions.
- Model comparison will use the time-aware, patient-grouped split described above as the primary validation strategy, with cross-validation restricted to the training partition only.
- Feature importance / SHAP values will be used to check which predictors drive the model's decisions.
- Given the dataset's modest size and the further reduction caused by the time-aware, patient-grouped split, performance metrics will be reported with a note on stability across folds (e.g., a range or standard deviation across cross-validation folds within the training partition) rather than as a single point estimate, since a smaller effective sample makes single-run metrics less reliable.

Determining whether the question is answered and the hypothesis is supported

- The research question is considered answered if the trained classifier outperforms the naive majority-class baseline by a meaningful margin on AUC-ROC and recall for the readmitted class, and the model can be served through a live prediction endpoint — demonstrating both predictive value and deployability.
- The hypothesis is considered supported if feature-importance or SHAP analysis shows utilization/complexity variables (time in hospital, number of prior inpatient visits, number of medications) ranking among the top predictors, consistent with the reasoning that care complexity drives readmission risk.

Timeframe (~6 weeks)

- Phase 1 (required, weeks 1-3): clean and engineer features; run clustering and PCA as exploratory unsupervised analysis; train the supervised classifier as a reusable, callable Python script using the time-aware/grouped split; log runs to MLflow; manually promote the best run to "Production"; build a FastAPI endpoint that serves predictions from the current Production model. This alone is a complete, defensible capstone.
- Phase 2 (stretch, weeks 4-6): wrap the Phase 1 training script in an Airflow DAG that retrains on a schedule, evaluates the new model against the current Production model using the same held-out validation logic, and conditionally promotes it only if it wins. FastAPI requires no changes, since it already reads whatever model is tagged Production. Environment setup for Airflow is the most likely place to lose time and has extra buffer budgeted for it.

Programming language and tools

- Language: Python.
- Libraries/tools: pandas and scikit-learn for data processing and baseline modeling; XGBoost for the primary supervised classifier; MLflow for experiment tracking and the model registry; FastAPI to serve predictions from whichever model is tagged "Production"; Airflow with docker-compose (stretch phase) to orchestrate scheduled retraining, evaluation, and conditional promotion; JupyterLab as the development environment.
- GitHub repository: https://github.com/AngieLie/AngelineSetiawan-hospital-readmission-risk-predictor

Data Dictionary

The table below lists key variables at this stage of the project. Engineered variables (marked as such) do not exist in the raw download and will be constructed during preprocessing to support the time-aware validation strategy described above. readmitted_30d is the target variable.

Variable Name	Description	Data Type	Role
encounter_id	Unique identifier for a single inpatient encounter.	Integer (ID)	Identifier / row key

Variable Name	Description	Data Type	Role
patient_nbr	Unique identifier for a patient; repeats across a patient's multiple encounters.	Integer (ID)	Grouping variable (used for patient-level train/test grouping)
race	Patient's recorded race/ethnicity category.	Categorical	Predictor
gender	Patient's recorded gender.	Categorical	Predictor
age	Ten-year age bracket (e.g., [50-60)).	Ordinal categorical	Predictor
admission_type_id	Coded category describing how the admission occurred (e.g., emergency, elective, urgent).	Categorical (coded)	Predictor
discharge_disposition_id	Coded category describing discharge destination (e.g., home, transferred, hospice).	Categorical (coded)	Predictor
admission_source_id	Coded category describing the referral source for admission.	Categorical (coded)	Predictor
time_in_hospital	Length of stay in days for the encounter.	Integer	Predictor (core to hypothesis)
num_lab_procedures	Number of lab tests performed during the encounter.	Integer	Predictor
num_procedures	Number of non-lab procedures performed during the encounter.	Integer	Predictor
num_medications	Number of distinct medications administered during the encounter.	Integer	Predictor (core to hypothesis)
number_outpatient	Number of outpatient visits in the year preceding the encounter (already lagged).	Integer	Predictor (core to hypothesis)
number_emergency	Number of emergency visits in the year preceding the encounter (already lagged).	Integer	Predictor (core to hypothesis)
number_inpatient	Number of inpatient visits in the year preceding the encounter (already lagged).	Integer	Predictor (core to hypothesis)
diag_1 / diag_2 / diag_3	Primary, secondary, and tertiary ICD-9 diagnosis codes.	Categorical (high-cardinality, coded)	Predictor (to be grouped/encoded)
number_diagnoses	Total number of diagnoses recorded for the encounter.	Integer	Predictor

Variable Name	Description	Data Type	Role
max_glu_serum / A1Cresult	Glucose serum and HbA1c test results, if performed.	Categorical	Predictor
change / diabetesMed	Whether diabetic medication was changed, and whether any diabetic medication was prescribed.	Binary categorical	Predictor
encounter_order (engineered)	Ordinal rank of an encounter based on encounter_id, used as a chronology proxy.	Integer (engineered)	Time-aware split key
days_since_prior_encounter (engineered)	Gap, in encounter order, since the same patient's previous recorded encounter.	Integer (engineered)	Predictor (lag/elapsed-time feature)
readmitted	Native three-class label: readmitted in <30 days, readmitted in >30 days, or not readmitted.	Categorical	Source of target variable
readmitted_30d (engineered)	Binary collapse of 'readmitted': 1 if readmitted within 30 days, 0 otherwise.	Binary integer (0/1)	TARGET

References

Agency for Healthcare Research and Quality. (2024). Nationwide Readmissions Database (NRD) overview. Healthcare Cost and Utilization Project (HCUP). https://hcup-us.ahrq.gov/nrdoverview.jsp

Centers for Medicare & Medicaid Services. (n.d.). Hospital Readmissions Reduction Program (HRRP). CMS.gov. https://www.cms.gov/medicare/payment/prospective-payment-systems/acute-inpatient-pps/hospital-readmissions-reduction-program-hrrp

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